Tribhuvan Univesity

Institute of Science and Technology

2072

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Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology Stat. 108
(Statistics I)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
All notations have the usual meanings.

Group A

Attempt any Two: (2×10=20)

1. Explain stratified random sampling method and also derive the properties of the estimate $\bar{y}_{st}$. In a survey, the following information was available from five strata.

No. of stratum	Total no. of sample (N_h)	No. of sample unit (n_h)	Sample mean ($\bar{y}_h$)	Sample variance (s_h^2)
1	80	7	2.2	0.5
2	95	9	2.7	0.6
3	105	11	3.1	0.4
4	115	13	3.7	0.4
5	125	14	3.9	0.3

With the help of the above information. Compute the sample estimate of population mean and population total, the variance of the estimate of sample mean and estimate of the total variance, and also to estimate the standard error of $\bar{y}_{st}$.

2. Describe the Kruskal Wallis one way analysis of variance (ANOVA) test.
A survey was conducted in order to know the differences in the level of morale among telephone operators of tele-companies. Sixteen telephone operators were selected at random from the three tele-companies and their morale level was recorded using standard scale and the standard scores so computed for each company are given below.

Telecom companies	NCEL	60	72	76	80	78	87
	NTC	55	60	65	70	75	
	UTL	67	59	56	65	70	

With the above information, use the Kruskal Wallis one way ANOVA test at 5% level of significance to test the hypothesis that the morale level of the telephone operators of three tele-companies is the same.

3. The following information shows the income and expenditure on food items of five families in a city.

Family	1	2	3	4	5
Expenditure on food (Rs.): Y	8000	3000	7000	4000	5000
Income (Rs): X_1	20000	10000	16000	12000	14000
Family members (size in number): X_2	7	3	6	3	5

a) Fit the linear regression model of Y on X_1 and X_2.
b) Estimate expenditure on food of a family with annual income Rs. 25000 and having 4 members.
c) Calculate the multiple determination and standard error of the estimate.

4+5
4 9/2
6+7
6 13/2

Group B

Answer any eight questions: (8×5=40)

4. Researcher selected 8 households from a cluster of 80 households using simple random sampling method and the number of persons observed for the selected households are found to be 3, 5, 7, 9, 6, 8, 10 and 6. Estimate the total population in the cluster, and also find its 95% confidence interval.

5. Describe in detail multistage sampling method with examples.

6. The test scores of two examinations of the college students are given in the following table.

Examination	Test score					
First term	57	58	56	49	63	69
Second term	56	45	59	59	51	67

With the help of the above information, to carry out the test statistic by using Wilcoxon matched pairs signed rank test at 5% level of significance that there is no statistical difference between the first and second test examinations scores of the college students.

7. Describe in detail the method of Kolmogorov Smirnov test in case of one sample.

8. Explain the method of the test of goodness of fit and test for independence with examples.

9. Multiple regression analysis model provided the following ANOVA table.

Source of variation	Sum of square	df
Regression variation	76854	2
Residual variation	81	10

(i) What is the sample size? And how many independent variables are in the regression model?
(ii) Compute R^2 and its adjusted value.
(iii) Carry out the overall goodness of fit test of the model at 5% level of significance.

10. Discuss autoregressive model for first order, and also explain problems involved in estimating the model.

11. How do you detect the presence of autocorrelation in the regression models?

12. Explain the properties of multiple correlation coefficients. It is observed that the simple correlation coefficients are found to be : $r_{12} = 0.51$, $r_{13} = 0.67$ and $r_{23} = 0.54$. Find $R_{1\,23}$ and $R_{3\,12}$ and compute the coefficient of multiple determination.

13. Write short notes on the following:
(a) Consequences of heteroscedasticity
(b) Sign test